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GROUNDHOG SHALLOW FOUNDATION ANALYSIS
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INPUT SUMMARY
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   Foundation geometry      : Rectangle
   Foundation depth, D      : 0.0000 m
   Skirted                  : No
   Foundation length, L     : 8.0000 m
   Foundation width, B      : 5.0000 m
   Length eccentricity, eL  : 0.0000 m
   Width eccentricity, eB   : 0.0000 m
   Drainage condition       : Drained
   Parameter source         : PCPT correlations
   Effective unit weight, γ′: 6.9745 kN/m³
   Friction angle, φ′       : 45.4794 deg
   Interface angle, δ′      : 40.4794 deg
   Effective stress, σ′v0   : 15.5000 kPa
   Bearing factor, RFb      : 2.0000
   Sliding factor, RFs      : 1.5000
   PCPT source workbook     : cpt.xlsx
   PCPT interval, zPCPT     : [1.00, 3.00] m
   PCPT cone factor, Nk     : 15.0000
   PCPT statistic           : Median

1. ANALYSIS BASIS
   Drainage condition       : Drained
   Parameter source         : PCPT correlations
   Foundation geometry      : Rectangle
   D [m]                    : 0.0000
   Skirted                  : No
   L x B [m]                : 8.0000 x 5.0000
   eL x eB [m]              : 0.0000 x 0.0000
   L' x B' [m]              : 8.0000 x 5.0000
     L' = L - 2eL = 8.0000 - 2 x 0.0000 = 8.0000 m
     B' = B - 2eB = 5.0000 - 2 x 0.0000 = 5.0000 m
     A = L x B = 8.0000 x 5.0000 = 40.0000 m^2
     A' = L' x B' = 8.0000 x 5.0000 = 40.0000 m^2
   A [m²]                   : 40.0000
   A' [m²]                  : 40.0000

2. SOIL INPUTS
   γ′ [kN/m³]               : 6.9745
   φ′ [deg]                 : 45.4794
   δ [deg]                  : 40.4794
   σ′v0 [kPa]               : 15.5000
   Bearing model            : API drained Nq-Nγ formulation with Kq and Kγ

2A. PCPT PARAMETER DERIVATION
   zPCPT = [zfrom, zto]      : [1.0000, 3.0000] m
   Rows in interval         : 101
   Adopted statistic        : Median
   Cone area ratio, a       : 0.8000
   Mapping unit weight      : 18.0000 kN/m^3
   Water level              : 0.0000 m
   Water unit weight        : 10.2500 kN/m^3
   Cone factor, Nk          : 15.0000

   Processing equations and assumptions
   qt = qc + u2(1-a)
   qnet = qt - σv0/1000                          [MPa]
   σ′v0 = σv0 - hydrostatic pore pressure  [kPa]
   φ′ = 17.6 + 11 log10[(1000 qt/Pa)/sqrt(σ′v0/Pa)]
          with Pa = 100 kPa
   su = 1000 qnet/Nk                                  [kPa]
   γ = 1.95 γw (1000 ft/Pa)^0.06
           x (σ′v0/Pa)^0.06                     [kN/m^3]
   γ′ = max(γ - γw, 0)                  [kN/m^3]
   δ′ = max(φ′ - 5 degrees, 0)
   ft is pressure-corrected sleeve friction where sleeve geometry
   is available; otherwise ft = fs.

   Interval statistics (valid n; min / mean / median / max)
   φ′ [deg]              : n=101   37.4221 / 44.5632 / 45.4794 / 46.5528
   su [kPa]              : n=101   198.8480 / 767.6611 / 850.7240 / 948.3067
   γ [kN/m^3]            : n=101   16.1248 / 17.1621 / 17.2245 / 17.5181
   σ′v0 [kPa]            : n=101   7.7500 / 15.5000 / 15.5000 / 23.2500

   Intermediate PCPT statistics (valid n; min / mean / median / max)
   qt [MPa]              : n=101   3.0360 / 11.5509 / 12.7972 / 14.2649
   qnet [MPa]            : n=101   2.9827 / 11.5149 / 12.7609 / 14.2246
   ft [MPa]              : n=101   0.0120 / 0.0561 / 0.0551 / 0.0880

   su-depth linear regression
   Valid samples            : 101
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   su(z) = slope x z + intercept
   Slope                    : -198.1805 kPa/m
   Intercept                : 1,164.0221 kPa
   R^2                      : 0.2995

   Adopted foundation parameters
   φ′                       : 45.4794 deg
   su                       : 850.7240 kPa
   su increase              : -198.1805 kPa/m (foundation input = 0.0000)
   γ                        : 17.2245 kN/m^3
   γ′                       : 6.9745 kN/m^3
   σ′v0                     : 15.5000 kPa
   δ′                       : 40.4794 deg

3. RESISTANCE FACTORS
   RFb [-]: 2.0000
   RFs [-]: 1.5000

4. SYMBOLS AND UNITS
   B [m]                 : Foundation width or diameter.
   R [m]                 : Foundation radius.
   L [m]                 : Foundation length.
   e, eL, eB [m]         : Applied load eccentricities.
   B' [m]                : Effective foundation width.
   L' [m]                : Effective foundation length.
   A [m^2]               : Full foundation area.
   A' [m^2]              : Effective foundation area.
   D [m]                 : Foundation embedment depth.
   qu [kPa]              : Ultimate bearing pressure.
   qu,f [kPa]            : Factored bearing pressure.
   Vult [kN]             : Ultimate vertical capacity.
   Vult,f [kN]           : Factored vertical capacity.
   Vb [kN]               : Bearing component of vertical capacity.
   Vadd [kN]             : Additional vertical-capacity contribution.
   V [kN]                : Vertical load or resistance.
   H [kN]                : Horizontal load or resistance.
   Hmax [kN]             : Maximum horizontal resistance on the V-H envelope.
   Hmax,f [kN]           : Factored maximum horizontal resistance.
   theta [deg]           : Load inclination used for an envelope point.
   Hb,outside [kN]       : Base sliding resistance outside the effective area.
   Aemb [m^2]            : Embedded vertical section area.
   Hb [kN]               : Base contribution to sliding resistance.
   Hs [kN]               : Skirt contribution to sliding resistance.
   Hu [kN]               : Ultimate sliding resistance.
   Hu,f [kN]             : Factored ultimate sliding resistance.
   RFb [-]               : Bearing resistance factor.
   RFs [-]               : Sliding resistance factor.
   zfrom, zto [m]        : Lower and upper depth boundaries of the selected PCPT data.
   zPCPT [m]             : Selected PCPT depth interval [zfrom, zto].
   γ′ [kN/m^3]          : Effective soil unit weight.
   φ′ [deg]              : Effective friction angle.
   σ′v0 [kPa]            : Effective vertical stress at foundation base.
   Nq [-]                : Effective-overburden bearing-capacity factor.
   Nq* [-]               : Nq term used in the bearing-pressure equation.
   Nγ [-]                : Unit-weight bearing-capacity factor.
   Kq [-]                : Combined correction factor for the Nq term.
   Kγ [-]                : Combined correction factor for the Nγ term.
   qv [kPa]              : Effective-stress contribution to qu.
   qγ [kPa]              : Unit-weight contribution to qu.
   sq, dq, iq, bq, gq [-]: Shape, depth, load, foundation and ground factors.
   sγ, dγ, iγ, bγ, gγ [-]: Corresponding unit-weight factors.
   δ [deg]               : Foundation-soil interface friction angle.
   Kp [-]                : Passive earth-pressure coefficient.
   Krd [-]               : Passive-resistance coefficient.

5. PRINCIPAL RESULTS
   qu [kPa]: 6,125.9768
     qu [kPa] = σ′v0 [kPa] x Nq* [-] x Kq [-]
                + 0.5 x γ′ [kN/m3] x B' [m]
                x Nγ [-] x Kγ [-]
       φ′ [deg] = 45.4794
       B' [m] / L' [m] = 5.0000/8.0000 = 0.6250
       Nq [-] = exp(pi x tan(φ′)) x tan^2(45 deg + φ′/2)
          = exp(pi x tan(45.4794)) x tan^2(45 + 45.4794/2) = 145.6377
       Nq* [-] = 145.6377 - 1.0000 = 144.6377
       Nγ [-] = 1.5 x (Nq [-] - 1) x tan(φ′)
              = 1.5 x (145.6377 - 1) x tan(45.4794) = 220.6179
       Kq [-] = iq [-] x sq [-] x dq [-] x bq [-] x gq [-]
          = 1.0000 x 1.4456 x 1.0000 x 1.0000 x 1.0000 = 1.4456
       Kγ [-] = iγ [-] x sγ [-] x dγ [-]
                      x bγ [-] x gγ [-]
              = 1.0000 x 0.7500 x 1.0000 x 1.0000 x 1.0000 = 0.7500
       qv [kPa] = 15.5000 x 144.6377 x 1.4456 = 3,240.9229 kPa
       qγ [kPa] = 0.5 x 6.9745 x 5.0000 x 220.6179 x 0.7500 = 2,885.0539 kPa
       qu [kPa] = 3,240.9229 + 2,885.0539 = 6,125.9768 kPa
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qu,f [kPa]: 3,062.9884
  qu,f = qu / RFb = 6,125.9768 / 2.0000 = 3,062.9884 kPa
Vult [kN]: 245,039.0708
  Vult [kN] = qu [kPa] x B' [m] x L' [m] + Vadd [kN]
    Vb [kN] = 6,125.9768 x 5.0000 x 8.0000 = 245,039.0708 kN
    Vadd [kN] = 0.0000 (no additional base-stress contribution)
    Vult [kN] = 245,039.0708 + 0.0000 = 245,039.0708 kN
Vult,f [kN]: 122,519.5354
  Vult,f = Vult / RFb = 245,039.0708 / 2.0000 = 122,519.5354 kN

Analysis of principal results
- Effective area retained: 100.00% of full area.
  A'/A x 100 = 40.0000 / 40.0000 x 100 = 100.00%.
- Entered eccentricities cause no effective-area reduction.
- Bearing factor reduces vertical resistance by 50.00%.
- Maximum V: Vult=245,039.07 kN unfactored and Vult,f=122,519.54 kN factored (50.00% reduction).
  Vult,f = Vult / RFb = 245,039.0708 / 2.0000 = 122,519.5354 kN.
  Reduction = (1 - Vult,f/Vult) x 100 = (1 - 122,519.5354 / 245,039.0708) x 100 = 50.00%.
- Maximum H: Hmax=27,553.72 kN unfactored and Hmax,f=18,369.15 kN factored (33.33% reduction).
  Hmax,f = Hmax / RFs = 27,553.7208 / 1.5000 = 18,369.1472 kN.
  Reduction = (1 - Hmax,f/Hmax) x 100 = (1 - 18,369.1472 / 27,553.7208) x 100 = 33.33%.
  For each positive envelope point i:
    Hi = Vi x tan(theta_i) + Hb,outside,i + Hs,i
  Hmax = max(Hi)
    = max(envelope H coordinates) = 27,553.7208 kN
  Hmax is a bearing-envelope coordinate; it is not the load-dependent Hu(V)
  described in Section 7. The sliding limit must
  also be satisfied at the relevant vertical load.
- qu/base effective stress = 6,125.9768 / 15.5000 = 395.224; Nq, Nγ and correction factors amplify resistance.
- Loads inside the factored envelope have available calculated resistance;
  loads outside the envelope exceed it. For drained loading, the factored
  sliding cutoff must also be satisfied.
- No applied design loads were entered, so utilization and pass/fail
  cannot be determined.

6. BEARING-CAPACITY COMPONENTS

qu [kPa]	: 6,125.9768
vertical_capacity [kN]	: 245,039.0708
N_q [-]	: 145.6377
N_γ [-]	: 220.6179
K_q [-]	: 1.4456
K_γ [-]	: 0.7500
s_q [-]	: 1.4456
s_γ [-]	: 0.7500
d_q [-]	: 1.0000
d_γ [-]	: 1.0000
i_q [-]	: 1.0000
i_γ [-]	: 1.0000
b_q [-]	: 1.0000
b_γ [-]	: 1.0000
g_q [-]	: 1.0000
g_γ [-]	: 1.0000

Analysis of Section 6 components
- For the principal vertical-capacity point, theta = nu = beta = 0 deg.
- iq = [1 - 0.5 tan(theta)]^5 = [1 - 0.5 tan(0)]^5 = 1.0000.
- iγ = [1 - 0.7 tan(theta)]^5 = [1 - 0.7 tan(0)]^5 = 1.0000.
- sq = 1 + iq(B'/L')sin(φ′) = 1 + 1.0000 x (5.0000/8.0000) x sin(45.4794) = 1.4456.
- sγ = 1 - 0.4iγ(B'/L') = 1 - 0.4 x 1.0000 x (5.0000/8.0000) = 0.7500.
- dq = 1 + 1.2(D/B')tan(φ′)[1-sin(φ′)]^2
  = 1 + 1.2 x (0.0000/5.0000) x tan(45.4794) x [1-sin(45.4794)]^2 = 1.0000.
- dγ = 1.0000 = 1.0000.
- bq = exp[-2 nu tan(φ′)] = exp[-2 x 0 x tan(45.4794)] = 1.0000.
- bγ = exp[-2.7 nu tan(φ′)] = 1.0000.
- gq = gγ = [1 - 0.5 tan(beta)]^5 = [1 - 0.5 tan(0)]^5 = 1.0000.
- N_q [-] = 145.6377 represents the effective-overburden
  contribution controlled mainly by effective_friction_angle [deg].
- N_γ [-] = 220.6179 represents the soil-unit-weight
  contribution beneath the effective foundation width.
- s_q [-] (shape factor) = 1.4456 increases the associated resistance term.
- d_q [-] (depth factor) = 1.0000 has a neutral effect.
- i_q [-] (load-inclination factor) = 1.0000 has a neutral effect.
- b_q [-] (foundation-inclination factor) = 1.0000 has a neutral effect.
- g_q [-] (ground-inclination factor) = 1.0000 has a neutral effect.
- K_q [-] = 1.4456 is the product of the five q factors above.
- s_γ [-] (shape factor) = 0.7500 reduces the associated resistance term.
- d_γ [-] (depth factor) = 1.0000 has a neutral effect.
- i_γ [-] (load-inclination factor) = 1.0000 has a neutral effect.
- b_γ [-] (foundation-inclination factor) = 1.0000 has a neutral effect.
- g_γ [-] (ground-inclination factor) = 1.0000 has a neutral effect.
- K_γ [-] = 0.7500 is the product of the five γ factors.
- qu [kPa] = 6,125.9768 is the resulting ultimate bearing pressure.
- vertical_capacity [kN] = 245,039.0708 is the
  corresponding vertical resistance over the effective dimensions, including

any additional base-stress contribution implemented by the method.

7. SLIDING-CAPACITY COMPONENTS

skirt_capacity [kN]	: 0.0000
K_rd [-]	: 4.4881
K_p [-]	: 4.7008

Analysis and derivation of Section 7 components
- No arbitrary fixed vertical load is assumed. Sliding resistance is evaluated as a function of the applied vertical load V.
- δ [deg] = 40.4794 (drained input).
- Hb(V) [kN] = V [kN] x tan(δ)
  = V x tan(40.4794) = 0.8535 x V.
- Kp [-] = tan^2(45 deg + δ/2)
  = tan^2(45 + 40.4794/2) = 4.7008.
- Krd [-] = Kp [-] - 1/Kp [-]
  = 4.7008 - 1/4.7008 = 4.4881.
- Aemb [m2] = D [m] x out-of-plane dimension [m] = 0.0000 x 8.0000 = 0.0000 m2.
- Hs [kN] = 0.5 x Krd [-] x γ′ [kN/m3] x D [m] x Aemb [m2]
  = 0.5 x 4.4881 x 6.9745 x 0.0000 x 0.0000 = 0.0000 kN.
- Hu(V) [kN] = Hb(V) + Hs = 0.8535 x V + 0.0000.
- Hu,f(V) = Hu(V) / RFs = (0.8535 x V + 0.0000) / 1.5.
- The dashed graph lines plot these equations over the full V range.
- The sliding resistance factor 1.5 is applied to the factored V-H envelope.

8. FORMULA SOURCES AND STANDARDS

FORMULATION BASIS
- API Recommended Practice 2GEO/ISO 19901-4:2003, Geotechnical and Foundation Design Considerations, 1st Edition, April 2011.
- API RP 2GEO is a modified U.S. national adoption of ISO 19901-4:2003. API issued Addendum 1 in October 2014.
- The Groundhog functions used by this application cite API RP 2GEO (2011) for the drained and undrained shallow-foundation bearing capacity, sliding resistance, effective-area and V-H failure-envelope formulations.

CURRENT ISO EDITION
- ISO 19901-4:2025, Oil and gas industries including lower carbon energy — Specific requirements for offshore structures — Part 4: Geotechnical design considerations, 3rd Edition, February 2025.
- ISO 19901-4:2025 supersedes ISO 19901-4:2016; the earlier 2003 and 2016 editions are withdrawn by ISO.
- The equations in this application retain the API RP 2GEO (2011) implementation. They have not been checked equation-by-equation against ISO 19901-4:2025 and must not be represented as compliance with, or verification to, the 2025 edition.

PCPT PARAMETER CORRELATIONS (WHEN PCPT INPUT IS SELECTED)
- Friction angle: Kulhawy, F.H. and Mayne, P.W. (1990), Manual on Estimating Soil Properties for Foundation Design, EPRI EL-6800.
- Undrained shear strength: Rad, N.S. and Lunne, T. (1988), Direct Correlations between Piezocone Test Results and Undrained Shear Strength of Clay, ISOPT-1, Vol. 2, pp. 911-917.
- Soil unit weight: Mayne, P.W., Peuchen, J. and Bouwmeester, D. (2010), Soil Unit Weight Estimation from CPTs, CPT'10, Paper 2-5.
- These are empirical parameter correlations, not foundation-design standards. Their applicability and parameter selection require engineering judgement and site-specific validation.

DESIGN-USE LIMITATION
- This report documents the calculation basis; it is not a declaration of code compliance. Confirm the governing edition, national or project requirements, load combinations, resistance factors and soil parameters before using the results for design.

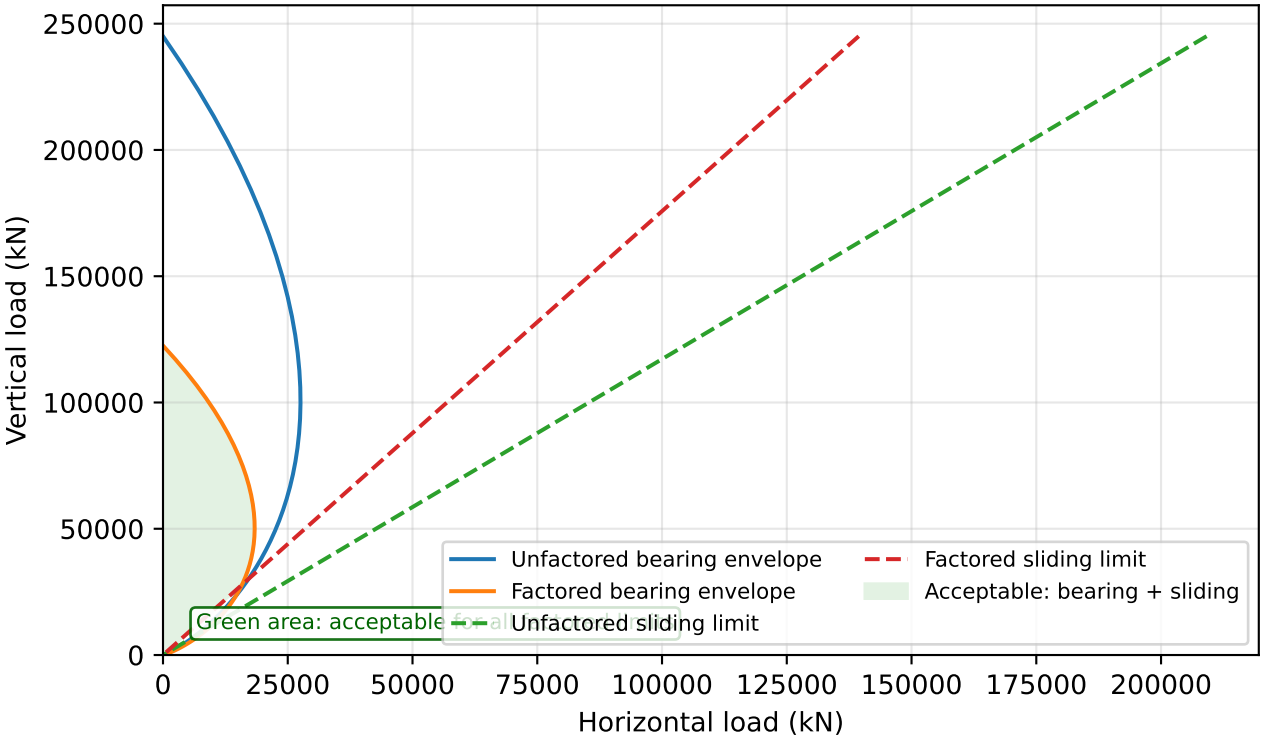
9. DESIGN NOTE

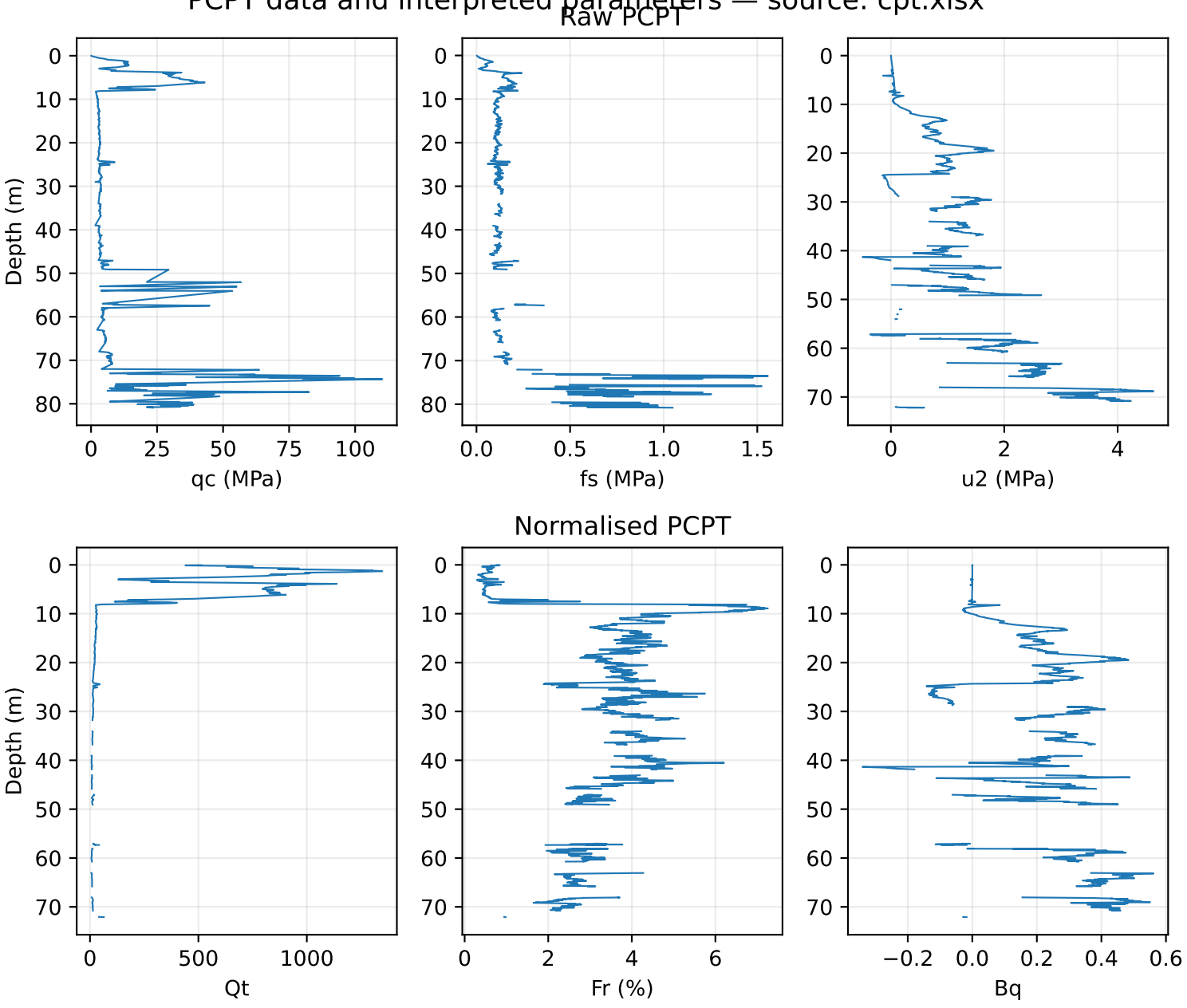
The undrained shear strength is an input soil parameter; it is not the bearing pressure. Bearing pressure includes capacity and geometry factors. Confirm correlation choices and resistance factors against the governing design standard and site-specific ground investigation.

10. CALCULATION RESULTS SUMMARY

Input φ′ [deg]	: 45.4794
Ultimate bearing pressure, qu [kPa]	: 6,125.98
Factored bearing pressure, qu,f [kPa]	: 3,062.99
Ultimate vertical capacity, Vult [kN]	: 245,039.07
Factored vertical capacity, Vult,f [kN]	: 122,519.54
Maximum envelope resistance, Hmax [kN]	: 27,553.72
Factored maximum resistance, Hmax,f [kN]	: 18,369.15
Effective foundation area, A' [m^2]	: 40.0000

These are calculated resistances. No applied design loads were entered, so utilization and pass/fail cannot be determined.


Drained shallow-foundation V-H capacity envelope
Vertical load (kN)
Horizontal load (kN)
0
50000
100000
150000
200000
250000
0
25000
50000
75000
100000
125000
150000
175000
200000
Unfactored bearing envelope
Factored bearing envelope
Unfactored sliding limit
Factored sliding limit
Acceptable: bearing + sliding
Green area: acceptable for all


PCPT data and interpreted parameters — source: cpt.xlsx
Raw PCPT
Depth (m)
qc (MPa)
fs (MPa)
u2 (MPa)
Normalised PCPT
Qt
Fr (%)
Bq